SEVeros Feature Matching System

In the technologically advanced city of Severos, the students of Millennia Science School are developing a new object detection system capable of identifying and tracking targets in complex, real world environments by combining advanced **feature detection** and **feature matching** techniques, allowing it to remain accurate despite variations in angle, motion, and lighting; to ensure its robustness, they train and evaluate the system using datasets composed of familiar student faces from multiple academies across Severos, simulating realistic and dynamic scenarios. Final output of the program is to display the most similar match for any detected face based on feature correspondence

- **Dataset Description**

  The datasets consist of:

  - **Target Object Image**

    A reference image representing the object or face to be detected.
  - **Student Images**

    A collection of images containing several Severos students used as candidates for matching.

- **Dataset Preprocessing**

  All images, target and object, need preprocessing steps such as **grayscale conversion** and **blur filter** to reduce noise and ensure consistency of feature extraction and matching. You can use **Gaussian Blur** with the following specifications:

  - Kernel size width: 3
  - Kernel size height: 3
  - SigmaX: 0

- Feature Detection

  In this stage, several feature detection methods such as **SIFT**, **AKAZE**, and **ORB** are used to extract **keypoints** and **descriptors**, with particular attention to the fact that **each method may produce different descriptor data types** (e.g., floating-point or binary), which **must be properly converted** to ensure compatibility with the FLANN.

- **Feature Matching**

  The target object image will be matched with pictures from the object images using the **FLANN** algorithm. Matching keypoints are filtered using **Lowe's Ratio Test** to eliminate unreliable correspondences, and different Lowe's ratio keypoints are evaluated to **determine the best match result**. The final output should be displayed **in color (RGB)** and **include all the lines** representing the matching keypoints.

Best Match Result: 1.7565872020075282